Highmark Website - Developer Guide

Code, architecture and tooling of the AEM Edge Delivery Services site

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Quick Reference

This guide is for the developers who maintain and extend the Highmark website on AEM Edge Delivery Services: its code, the content structures the code depends on, and the tooling that builds the site's generated content. The companion **Author Guide** covers how content is authored.

Project URLs

Resource	URL
Code repository	`https://github.com/AdobeDrago/highmark` (branch `main`)
Preview	`https://main--highmark--adobedrago.aem.page/`
Live	`https://main--highmark--adobedrago.aem.live/`
Branch preview	`https://<branch>--highmark--adobedrago.aem.page/`
Local development	`http://localhost:3000/` (`aem up`)
Content (Document Authoring)	`https://da.live/#/adobedrago/highmark/`
Content source	`https://content.da.live/adobedrago/highmark/`
Admin API	`https://admin.hlx.page/` (preview, publish, index, status, config)
DA Source API	`https://admin.da.live/` (read and write DA documents and sheets)

Site configuration (Config Service)

Setting	Value
Organization / site	`adobedrago` / `highmark`: a single, standard (not repoless) site
Code	GitHub `adobedrago/highmark`, synced by the AEM Code Sync GitHub app
Content	`https://content.da.live/adobedrago/highmark/` (type `markup`, Document Authoring)
Indexes	`query.yaml`: `main-index` (`/query-index.json`) and `search-index` (`/search-index.json`)
Access	Admin role only; sign-in required for preview and publish (`requireAuth: auto`)
Production CDN	None configured yet: `aem.live` is the production host
Sitemap, headers, robots	Not configured (defaults)

There is no `fstab.yaml` or `helix-config.yaml`; everything above lives in the Config Service (`https://admin.hlx.page/config/adobedrago/sites/highmark.json`).

Architecture Overview

Tech stack

- Vanilla JavaScript (ES modules) and CSS with custom properties. No build step, no framework, no runtime dependencies: files are served as committed.
- Content from Document Authoring (DA), rendered by the Edge Delivery Services pipeline into semantic HTML that the code decorates.
- Started on 2026-08-17 from Adobe's AEM Block Collection (DA flavour); `package.json` still carries its name.

Project structure

```
├── blocks/            40 block folders (29 project blocks, 11 untouched boilerplate)
├── scripts/
│   ├── aem.js         Core library (unmodified since the initial commit)
│   ├── scripts.js     Page decoration, loading phases, template and ZIP logic
│   ├── delayed.js     Empty
│   ├── sidekick.js    DA experimentation hook for the Sidekick
│   └── context.js     Unused DA context-editing helper
├── styles/
│   ├── styles.css     Global and section styles (imports brand.css)
│   ├── brand.css      Brand tokens: fonts, colours, heading sizes
│   ├── fonts.css      Local @font-face (loaded lazily)
│   ├── lazy-styles.css  Empty
│   └── templates/     34 page-template stylesheets
├── icons/             14 SVG icons (:name: syntax)
├── fonts/             Local font files (mostly unused, see Known Issues)
├── plugins/experimentation/  Vendored, never loaded
├── tools/             Migration and content-generation tooling (see "Tooling")
├── head.html          CSP, Adobe Fonts kit, aem.js, scripts.js, styles.css
└── .github/workflows/main.yaml   CI: lint on every push
```

There is no `templates/` folder: page templates are CSS files in `styles/templates/` plus a few body-class hooks in `scripts.js` (see "Templates").

Loading phases (eager, lazy, delayed)

Phase	What this site loads	Notes
Head	CSP meta, `use.typekit.net/iaj7yko.css`, `aem.js`, `scripts.js`, `styles.css`	The Adobe Fonts stylesheet is **render-blocking** in `head.html`. It was kept deliberately to match highmark.com's font metrics.
Eager (`loadEager`)	Template classes, every template stylesheet in `Template` metadata, `decorateMain`, the per-template decorators, the first section	`body.appear` waits for the template CSS, so a template stylesheet blocks first paint. `fonts.css` also loads eagerly on screens 900px and wider.

Phase	What this site loads	Notes
Lazy (`loadLazy`)	Modal link handler, remaining sections, hash scroll, header, breadcrumbs, footer, `lazy-styles.css`, fonts, ZIP tokens and modal	
Delayed (`loadDelayed`)	`delayed.js` after 3 seconds	The file is empty: there is no analytics, tag manager or consent tool. Only Adobe's built-in RUM runs (about 1 in 100 visits; force with `?rum=on`).

Rules: add third-party scripts to `delayed.js`, never to `scripts.js` or `head.html`. Keep the first section's blocks light; they load eagerly.

Content Security Policy. `head.html` sets `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none'` and moves it to an HTTP header. Scripts created from already-trusted scripts are allowed through `strict-dynamic`; a `<script>` tag added straight to `head.html` needs `nonce="aem"`.

`scripts/scripts.js`

It imports from `aem.js`: `buildBlock`, `decorateBlock`, `loadBlock`, `loadHeader`, `loadFooter`, `decorateButtons`, `decorateIcons`, `decorateBlocks`, `decorateTemplateAndTheme`, `getMetadata`, `waitForFirstImage`, `loadSection`, `loadSections`, `loadCSS`, `sampleRUM`, `readBlockConfig`, `toClassName` and `toCamelCase`. Its own `decorateSections` replaces the one in `aem.js`.

Function	Phase	What it does
`buildHeroBlock`, `buildAutoBlocks`	Eager	Auto-builds a `hero` when a picture precedes the H1 outside any block; skipped for template `medicare-hra`
`decorateMain` (exported)	Eager	Buttons, icons, auto blocks, sections, blocks. Also used by `fragment.js`.
`decorateShopHomeLinks`	Eager	`body.shop-home`: off-site and PDF links open in a new tab
`decorateProviderContent`	Eager	`body.provider-content`: classes the "Last Updated" line and the news article heading parts (`article-topic`, `-title`, `-regions`, `-date`, `-for`)
`decorateLeadershipBio`	Eager	`body.leadership-bio`: classes paragraphs by their text, and the LinkedIn link
`decorateLegalLinks`	Eager	`body.legal`: classes four CMS-ONC API links by their text
`decorateFaqLinks`	Eager	`body.faq`: links holding `:external-link:` open in a new tab, with screen-reader text
`loadEager`	Eager	`lang`, template classes and CSS, breadcrumb row reservation, decorators, first section, RUM
`autolinkModals`	Lazy	Click delegate: links containing `/modals/zip-county` open the ZIP modal; other `/modals/` links open `blocks/modal`
`autoOpenShopZipModal`	Lazy	Applies ZIP tokens when `Theme: shop` or tokens are present; auto-opens the ZIP modal on `Theme: shop` pages with no stored selection

Function	Phase	What it does
`loadBreadcrumbs`	Lazy	Builds the `breadcrumbs` block when `Breadcrumbs: true`
`loadLazy`, `loadDelayed`, `loadSidekick`, `loadPage`		Phase orchestration; `sidekick.js` on `sidekick-ready`
`da()`	Module	`?dapreview` loads DA's live preview; `?daexperiment` loads DA's experiment UI

Templates in code

- `TEMPLATE_STYLES` (`scripts.js`) lists the 34 template names that have a stylesheet; it matches `styles/templates/` exactly. `loadEager` splits the comma-separated `Template` metadata, normalises each value with `toClassName` and loads each listed stylesheet in parallel.
- Every `Template` and `Theme` value also becomes a body class (`decorateTemplateAndTheme`), listed or not. `chip`, `chip-landing`, `topic-hub-subnav` and `shop` are styled from `styles/styles.css` and block CSS instead of a template file.
- **Adding a template** takes a file in `styles/templates/` and an entry in `TEMPLATE_STYLES`.

Client storage, events and URL state

Kind	Name	Owner
localStorage	`shop-zip-county` (`{zipCode, county, state, regionCode, region}`)	`blocks/zip-modal/zip-store.js`
sessionStorage	`disabledAlertBanner` (highmark.com's own flag)	`blocks/alert-banner`
sessionStorage	`fonts-loaded`	`scripts.js`
Custom event	`zip-county-submit` (bubbles, `detail` = the selection)	Dispatched by `zip-county-form`, handled by the shop header
URL state	`?q=`	`blocks/search` (`replaceState`)
URL state	`?keyword=&group=&year=`	`blocks/press-release-list` (`pushState`)

Local Development

Prerequisites

- **Node.js 20.** CI uses 20. There is no `.nvmrc` or `engines` field yet; the tools need at least 18 (global `fetch`, `FormData`, `Blob`).
- **AEM CLI:** `npm install -g @adobe/aem-cli` (tested with 16.21).
- **Git with LF line endings.** `.gitattributes` sets `eol=lf` and ESLint enforces `linebreak-style: unix`. On Windows set VS Code's `files.eol` to `\n`.

Setup

```
git clone https://github.com/AdobeDrago/highmark.git
cd highmark
npm install
aem up            # http://localhost:3000, proxying preview content
```

- `aem up` serves the working tree's code with content from `main--highmark--adobedrago.aem.page`, and reloads on file changes. A single site needs no `--pages-url`.
- **Local-only HTML:** `aem up --html-folder content` also serves files from the gitignored `content/` folder at `/content/<path>`. When the header or footer fragment named in metadata is missing, they fall back to `/content/nav.plain.html` and `/content/footer.plain.html`.
- **For the test suites,** run `aem up --port 3100 --no-open` (their default `HOST`).

Linting

```
npm run lint        # eslint . && stylelint "blocks/**/*.css" "styles/**/*.css"
```

ESLint extends `airbnb-base` (browser env, `.js` extensions required in imports, `import/no-cycle` off). Stylelint uses `stylelint-config-standard`. CI runs the same command on every push.

Credentials

Nothing is stored in the repo.

- **Preview and publish** go through the AEM Sidekick or the DA editor, signed in with your Adobe account. Your account needs a role in the site's `access` config.
- **Admin API from scripts:** an `x-auth-token` from `https://admin.hlx.page/login/adobedrago/highmark/main`.
- **DA Source API and the tools in `tools/`:** an IMS token. `npx github:adobe-rnd/da-auth-helper token` writes `~/.aem/da-token.json` (about 24 hours); `aem content` writes `.hlx/.da-token.json` (gitignored). The tools read either file and send it as a Bearer token.

- **Windows / Git Bash:** prefix tool commands that pass `/paths` as arguments with `MSYS_NO_PATHCONV=1`, or MSYS rewrites them into Windows paths.

Design System

Tokens

Brand tokens are in `styles/brand.css`, imported at the top of `styles.css`.

```
/* fonts */
--heading-font-family: sofia-pro, "Helvetica Neue", helvetica, arial, sans-serif;
--body-font-family: roboto, "Helvetica Neue", helvetica, arial, sans-serif;

/* colours */
--background-color: #fff;
--text-color: #003963;        /* Highmark navy: headings and base text */
--text-body-color: #001446;   /* body copy */
--link-color: #0066b1;        /* Highmark blue */
--link-hover-color: #003963;
--light-color: #f7fbfd;       /* styles.css */
--dark-color: #003963;        /* styles.css */

/* section tints */
--highlight-color: #fef0ec;   /* blush */
--accent-color: #f7fbfd;      /* polar */
--light-blue-color: #d9ebf6;  /* splash */
--grey-color: #f1f4f4;        /* paper grey */

/* heading sizes: desktop (brand.css); below 900px styles.css overrides them */
--heading-size-xxl: 56px;  /* 36px below 900px */
--heading-size-xl: 34px;   /* 28px */
--heading-size-l: 24px;
--heading-size-m: 20px;
--heading-size-s: 18px;
--heading-size-xs: 16px;

/* body sizes */
--body-font-size-m: 16px;
--body-font-size-s: 16px;
--body-font-size-xs: 14px;

/* layout */
--content-max-width: 1200px;
--nav-height: 84px;           /* header bar below 992px */
--header-height: var(--nav-height);
--breadcrumbs-height: 44px;   /* 94px from 992px */
```

Most page layouts don’t use these tokens: they carry the source site’s measured values (px sizes, unitless line-heights) directly in the template and block CSS. That was deliberate, to match highmark.com within a pixel. Keep unitless values unitless: a px line-height such as 26.4px rounds differently in Chrome and shifts later images by a pixel.

Fonts

Family	Source	Weights	Usage
Sofia Pro	Adobe Fonts kit `iaj7yko` (`head.html`)	Kit weights	Headings, nav labels
Roboto	Adobe Fonts kit `iaj7yko`	Kit weights	Body text
Roboto Condensed	`fonts/roboto-condensed-bold.woff2` (`fonts.css`)	700	Declared, but no rule uses it
Material Icons	`fonts.gstatic.com` (from `cards-minimal-dark-iconnav.css`)		Ligature icons typed as text in the quick-link bar

Fallback faces `roboto-fallback` and `roboto-condensed-fallback` (Arial, size-adjusted) are in `styles.css`. Local Roboto files used to override the kit; their vertical metrics moved text, so they were dropped from `fonts.css`, and `fonts/roboto-{regular,medium,bold}.woff2` are now unused.

Breakpoints

The site follows highmark.com's grid rather than the boilerplate's:

Breakpoint	Media query	Usage
Phone	default, `width < 768px`	Phone layouts
Tablet	`width >= 768px` (about 160 rules)	Tablet layouts, 64px gutters
Desktop	`width >= 992px` (about 150 rules)	Desktop header, multi-column layouts
Boilerplate	`width >= 900px` (about 40 rules)	Older block rules and heading sizes
Wide	`width >= 1200px`, `>= 1366px`	Card grids (three 400px slots from 1366px)

Several one-off breakpoints (428, 576, 1024, 1100px) copy the source's art-direction and layout switches. Use 768 and 992 for new work.

Section styles

Global section styles (`styles.css`): `light`, `accent`, `light-blue`, `blue`, `highlight`, `highlight text-band`, `dark`, `center`, `grey`. Template-scoped styles are listed per template in the Author Guide. Section metadata reaches the page as classes and `data-` attributes on the section `div` (the pipeline converts it in `.plain.html`; `scripts.js` handles unprocessed HTML).

Header height reservation

`styles.css` reserves the header's height before it loads, keyed on the literal `nav` metadata value:

nav value	Below 992px	From 992px
default (`/nav`)	84px	196.5px
ends `/providers/fragments/nav`	84px	270.5px
starts `/member/fragments/`	84px (69px for `/member/fragments/bcbswny/nav`)	154px
ends `/shopx-header`	67px	208px
plus breadcrumbs row	44px	94px

A new header fragment path needs a rule here, or it gets the corporate reservation and the page shifts when the header loads.

Blocks Reference

Inventory

Block	Origin	Commits	Last change
accordion, carousel, columns, embed, form, fragment, modal, quote, table, tabs, video	Boilerplate, untouched	1	2026-08-17
cards	Boilerplate + `shop` variant	4	2026-10-07
hero	Boilerplate + CSS `light` variant	2	2026-09-10
header, footer, search	Boilerplate, rewritten	19, 16, 6	2026-10-19
hero-minimal-dark-withimg	Project	21	2026-10-19
columns-minimal-dark	Project	15	2026-10-16
cards-minimal-dark-list	Project	13	2026-10-16
cards-minimal-light-sidenav	Project	11	2026-10-16
cards-minimal-dark-iconnav, -withimg, -withimg-icons	Project	10 each	2026-10-05 to 10-19
cards-minimal-dark-withimg-2, form-minimal-light	Project	9 each	2026-10-05, 10-14
accordion-minimal-light	Project	6	2026-10-14
columns-minimal-dark-overlay, zip-county-form, zip-modal	Project	5 each	2026-10-02 to 10-19
press-release-list	Project	4	2026-10-06
breadcrumbs, table-minimal-dark-compare	Project	3 each	2026-10-08, 10-01
alert-banner, hero-minimal-light, provider-sidenav	Project	2 each	
enrollment-form, flip-cards, forms-library, provider-list, provider-panel	Project	1 each	2026-10-08 to 10-15

The untouched boilerplate blocks are unused on live pages, apart from `columns`, `table` and `fragment` (restyled by templates) and `modal` (used through links). The block READMEs and `metadata.json` files come from the importer and are partly stale (several say “No variations” for blocks that have them).

Project blocks

Block	Purpose	Variants (JS-checked in bold)	Key technical details
hero-minimal-dark-withimg	Page hero, photo banner	sub, navy, light, dark, **intercept**	Exports buildArtDirectedPicture and INTERCEPT_MEDIA. Three images in the first cell become one art-directed <picture> (switch at 428 / 768px; intercept at 768 / 992px). Largest block CSS; restyled by 13 templates.
hero-minimal-light	Title band, no image		CSS only (the JS is a no-op)
cards-minimal-dark-withimg	Image card grid	narrow, underline, mobile-center, compact, small-image (template-scoped CSS)	createOptimizedPicture at 750 and keeps authored width and height
cards-minimal-dark-withimg-2	Two wide promo cards		JS identical to boilerplate cards.js apart from the class prefix
cards-minimal-dark-withimg-icons	Illustrated cards	underline	Same JS as −2. underline is added by CSS context when in a center section without links.
cards-minimal-dark-list	Text cards		Adds an hr under each H3 (see Known Issues)
cards-minimal-dark-iconnav	Quick-link bar		Icon cell: image, :icon:, or a Material Icons ligature name as text
cards-minimal-light-sidenav	Left nav		Rows parent, current, sibling, child (legacy active). Two shapes: guide (any sibling) and article. Falls back to matching the current path when no row is current.
columns-minimal-dark	Text beside a photo		Adds -N-cols and -img-col; a cell holding only an .mp4 link becomes a <video>
columns-minimal-dark-overlay	Photo band with overlaid text		Imports the hero's picture builder; adds -overlay-right when the image cell is first; reads an ancestor .highlight section
accordion-minimal-light	FAQ accordion		Rows become details/summary; first item opens only on body.chip, .answers-landing, .faq
table-minimal-dark-compare	Comparison tables	**no-header**, data	Copy of table.js that drops the importer's artifact first row
alert-banner	Homepage notification		Moves itself before <header>; sessionStorage disabledAlertBanner; fixed id #alert-banner-title
breadcrumbs	Trail with Print and Share		Built by scripts.js, not authored. Fetches /query-index.json?limit=5000, then each ancestor's page HTML for its breadcrumb-

Block	Purpose	Variants (JS-checked in bold)	Key technical details
			`title`. `navigator.share` with a clipboard fallback.
search	`/search` results		Exports `loadIndex`, `searchIndex`, `queryTerms`, `pageTitle`, `fallbackSearchUrl` for the header. Fetches `/search-index.json?limit=5000`; ranking in four groups, provider pages last; `NON_PAGE` regex excludes non-pages.
press-release-list	Press releases, events	**`events`**	Multi-sheet (`data`, `locations`) at `?limit=1000`; filters in the URL (`pushState`, `popstate`); loads `/docs/library/icons/filter.svg` from content
provider-list	Provider article and document lists	**`documents`**	Sheet at `?limit=5000`; `Filters` row; region items tinted; PDF and lock icons from `/icons/`
provider-panel	Provider grey and region boxes	**`region`**, `plain`	Region badges as `ul[aria-label=Regions]`
provider-sidenav	Provider left menu		Fetches the fragment's raw `.plain.html` (not `loadFragment`), so `decorateMain` doesn't run on it; calls `decorateIcons` itself
forms-library	Regional Forms Library		Category and keyword filters; hides the section's intro paragraphs while filtering; hard-coded h2/h3
form-minimal-light	Contact, fraud, news-alert forms		Wraps `blocks/form/form.js` (doesn't load `form.css`). Adds floating labels, help and error text, and `other-of-<select>` conditional fields. POSTs JSON to the block's second link.
zip-county-form	Shop ZIP and county form		Uses `form-fields.js` and `zip-store`; writes `shop-zip-county`; dispatches `zip-county-submit`; closes its `<dialog>`. Standalone mode follows a same-origin `?redirect=`.
zip-modal	ZIP modal and region layer (helper, not authorable)		`zip-modal.js` (`openZipModal`, `autoOpenZipModal`), `zip-store.js` (storage, sheet paths, memoised `fetchSheet`), `zip-tokens.js` (tokens, `data-regions`, location line)
enrollment-form	Shop enrollment first step		Validates `effective-date` and `radio` fields, then navigates to the authored link. Nothing is posted or stored.
flip-cards	Shop life-event picker		Front and back swapped with `inert`; SVG icons kept as-is

Header and footer

Both pick a layout from the **shape of the fragment** named in `nav` / `footer` metadata (default `/nav`, `/footer`):

Fragment	Header layout	Footer layout
Any section styled `shop-*`	Shop header (`shop-header.js`)	Shop footer (`shop-footer.js`)
3 sections	Corporate: utility, brand, menu	
4 sections	Provider: corporate + a search-target link	
2 sections	Regional member: brand, menu (can inherit `/nav`'s menu)	
Brand, columns, legal		Corporate; `footer-regional` for `/member/fragments/`, `footer-providers` for `/providers/` paths

- **Sections are read by position**, not by name (`header.js`, `footer.js`). Reordering a fragment's sections breaks the layout.
- The menu section's `full width` style gives the ruled full-width dropdowns. `placeSearch` moves the one search form between the desktop nav row and the phone bar at 992px.
- The header search box lazy-loads `blocks/search/search.js` on first focus and uses the same index and ranking as `/search`. Pre-typing suggestions come from `/search-suggestions.json`.
- The shop header and footer re-apply ZIP tokens and `Regions` sections when the visitor changes ZIP (`zip-county-submit`).

ZIP and region layer

```
/modals/zip-county (DA fragment)
  └─ zip-county-form ── reads /shop/zip-county-form.json, /shop/zip-counties.json, /shop/
regions.json
        └─ writes localStorage shop-zip-county, dispatches zip-county-submit
zip-tokens.js (applyZipTokens)
  ├─ {{token}} in hrefs (pipeline percent-encodes braces; decoded here) and text
  ├─ hides a line until all its tokens are filled (.zip-token-line / .zip-token-ready)
  ├─ [data-regions] sections shown only for matching regionCode (or "none")
  ├─ .zip-location: "<County> County, <ST> <ZIP>" at the start of the first paragraph
  └─ filled off-site or PDF links get target=_blank
```

- Tokens are the selection's fields plus every `regions` sheet column, lowercased and hyphenated.
- `applyZipTokens` runs on page load only when `Theme: shop` or the page contains a token; otherwise `Regions` sections all show until a submit.
- A stored selection without `regionCode` (from an earlier version) counts as none.
- The ZIP data is interim, built from Census data (see "Tooling", `zip-data`).

Cross-block dependencies

Module	Used by
`blocks/fragment` (`loadFragment`)	modal, zip-modal, shop header and footer

Module	Used by
`blocks/modal` (`createModal`, `openModal`)	zip-modal, `scripts.js`
`blocks/form` (`form.js`, `form-fields.js`)	form-minimal-light, zip-county-form
`blocks/zip-modal` (`zip-store`, `zip-tokens`)	shop header and footer, zip-county-form, `scripts.js`
`blocks/search`	header
`blocks/hero-minimal-dark-withimg` (`buildArtDirectedPicture`)	columns-minimal-dark-overlay

Template stylesheets restyle the shared blocks heavily: `hero-minimal-dark-withimg` is restyled by 13 templates, `columns-minimal-dark` by 14. When you change a block's base CSS, check the templates that override it (search `styles/templates/` for the block class) and test their example pages.

Templates

Applied with `Template: <name>` in page metadata (comma-separated for several). Each stylesheet is scoped to `body.<name>`.

Template	Pages	Special behaviour
`home`	`/`	Homepage layout; alert banner; art-directed hero and bands
`about-article`, `leadership`, `leadership-bio`, `bright-blue-futures`, `events`	About section	`leadership-bio`: `decorateLeadershipBio` text-to-class mapping
`topic-hub-landing`, `topic-hub-subnav`, `spending-account`, `answers-landing`	`/resources`	`topic-hub-subnav` has no file (styled in `styles.css`)
`faq`	FAQ landing and topics	`decorateFaqLinks`; accordion opens first item; spacer section styles
`glossary`	Health Insurance Glossary	A–Z bar and letter groups
`legal`	Legal, fraud and privacy pages	`decorateLegalLinks`; link-to-button conventions differ
`member-regional`, `member-subpage`	`/member/<region>` and sub-pages	`member-subpage` only as a second value (`faq, member-subpage` / `legal, member-subpage`)
`chip`, `chip-landing`	Western PA CHIP	No files (styled in `styles.css`); accordion opens first item on `chip`
`state-plans`, `get-help`, `learn-about-medicare`, `blue-neighbors`	`/plans`	
`employer-landing`, `employer-subpage`	`/employer`	Some rules keyed on heading ids
`campaign-landing`, `subsidiary-home`	Campaign and subsidiary pages	Some rules keyed on heading ids
`contact`, `media-contacts`, `capitol-hill-report`, `news-alert`, `language-assistance`, `find-care`	Single pages	
`providers`, `provider-content`	`/providers`	`decorateProviderContent` (article heading parts)
`shop-home`, `shop-page`	`/shop`	`decorateShopHomeLinks`

Template	Pages	Special behaviour
`content-landing`, `zipcode-gate-login`	Preview only	Not yet published
`medicare-hra`	None live	Only disables the auto hero
`press-releases`	`/newsroom/press-releases`	No styles; the value does nothing

Content the Code Depends On

The code reads these DA paths at runtime. Renaming or moving one breaks the feature.

Path	Read by
`/nav`, `/footer`, `/member/fragments/<region>/{nav,footer,footer-basic}`	header, footer
`/shop/fragments/shopx-header[-dental|-blue-edge-balance]`, `/shop/fragments/shopx-footer`	shop header and footer
`/providers/fragments/{nav,footer,service-centers,spotlight}`, `/providers/fragments/sidenav/<section>`	header, footer, fragment, provider-sidenav
`/modals/zip-county`	zip-modal (hard-coded)
`/shop/zip-county-form.json`, `/shop/zip-counties.json`, `/shop/regions.json`	zip-store (defaults)
`/search-index.json`, `/query-index.json`, `/search-suggestions.json`	search, breadcrumbs, header
`/providers/lists/*.json`, `/newsroom/press-releases-data.json`, `/about/events-data.json`	provider-list, press-release-list (linked from the block)
`/docs/library/icons/filter.svg`	press-release-list
`/metadata.json` (bulk metadata)	pipeline: `nav`, `footer`, `theme`, `breadcrumb-home` for `/shop/**`, `/providers/**`, `/drafts/**`
`/redirects.json`	pipeline (generated by `tools/redirects`)

Indexes are defined in the Config Service's `query.yaml`, not in the repo:

- `main-index`: title, description, image and lastModified for every page; used by breadcrumbs and the redirects tool.
- `search-index`: title, description and the first 5,000 words of `main`. It excludes fragments, drafts, sheets, `/nav`, `/footer`, `/search`, `/modals/**`, `/docs/**` and `/shop/beta/**`.

After changing `query.yaml`, re-index with `POST https://admin.hlx.page/index/adobedrago/highmark/main/*`. The search index is about 1 MB (280 KB compressed) for about 285 pages and is downloaded whole on first use. A full migration (about 1,400 pages) would make it around 5 MB (1.4 MB compressed), so the full site needs a search service.

DA block library: the DA config's `library` lists Blocks (`/docs/library/blocks.json`), Templates (`/docs/library/templates.json`, which doesn't exist) and Icons. Each block doc's sections are Library items: a `library-metadata` block (`name`, `description`), then the block. Add an item when you add a block or variant.

Tooling

Everything under `tools/` is Node ESM (`node tools/<dir>/<script>.mjs`). Shared conventions:

- Constants: org `adobedrago`, site `highmark`, branch `main`.
- **Read-only by default.** Writes happen only behind `--upload` or `--apply`, and they write to DA and preview, **never publish** (except `provider-import/admin.mjs live`).
- Without a flag, the data builders write their JSON to the current directory: run them from a scratch directory and don't commit the output.
- Optional dependencies, not in `package.json`: `npm i --no-save playwright linkedom` and `npx playwright install chromium`.
- `tools/` isn't in `.hlxignore`, so committed files there are publicly served from the code bus at `/tools/...`.

Generated content and its tools

Tool	Builds	Command	Keep after migration?
`redirects/build-redirects.mjs`	`/redirects.json`	`node tools/redirects/build-redirects.mjs [--upload]`	Yes, until no internal link points at an unmigrated page
`provider-lists/build-provider-lists.mjs`	`/providers/lists/<list>.json` (5 lists)	`... [--upload]`	Yes, while provider news is a feed snapshot (weekly)
`press-releases/build-press-releases-data.mjs`	`/newsroom/press-releases-data.json`	`... [--upload]`	While highmark.com is the source of releases
`events/build-events-data.mjs`	`/about/events-data.json`	`... [--upload]`	While highmark.com is the source of events
`zip-data/build-zip-data.mjs`	`/shop/zip-counties.json`, `/shop/regions.json`	`... [--upload]`	Until enGen's ZIP list replaces it; update each plan year
`forms/build-*-form-fields.mjs`	Contact and fraud form field sheets	`... <importer json> [--upload]`	No (maintain the sheets in DA)

Redirects (critical)

`build-redirects.mjs` crawls every page in the live query index, plus the fragments in its `FRAGMENTS` list and the `Link` column of the sheets in its `LISTS` list. It finds links that 404 on live and writes a row sending each to the same path on its source (`/providers/*` to providers.highmark.com, the rest to highmark.com), or to our page when the source itself redirects there.

- **Any path with a DA document, published or not, gets no row.** An unpublished draft therefore 404s on live once the regenerated sheet is published. Drafts that should stay redirected, and retired URLs, go in `FORCE`.
- **Redirects beat pages:** a row hides any page published at its path. Publish a page and the regenerated sheet together.
- **The sheet is shared.** Before publishing, compare DA (`admin.da.live/source/adobedrago/highmark/redirects.json`) with live, and check that every dropped row's page is live or is being published with it.
- Rows also disappear when the source returns non-200 at run time. The run aborts without writing on any 429 or 5xx.
- Add new link-bearing fragments to `FRAGMENTS` and new list sheets to `LISTS`. Never hand-edit the sheet.
- A page published while a redirect covered its path isn't indexed: re-index it (`POST https://admin.hlx.page/index/adobedrago/highmark/main/<path>`).

Provider lists

Snapshots `providers.highmark.com/bin/prc/blog` and `/bin/prc/topics` into five sheets (`Title, Link, Type, Topic, For, Published, Updated, Regions, Description, Gated, Source`).

- Links become valid page paths (lowercase, anything outside `a-z0-9-` becomes `-`); `Source` keeps the original URL, which becomes the redirect destination.
- **Order:** re-run, publish the sheets, then rebuild and publish `/redirects`.
- A new list must also be added to `LISTS` in `build-redirects.mjs` and in `tools/tests/list-links.mjs`.

ZIP data

Builds the interim shop data from the Census 2020 ZIP-to-county file (downloaded each run) and county-to-region tables hard-coded from Highmark's service-area map.

- Drops counties with under 2% of a ZIP's land.
- PO-box ZIPs are missing; `EXTRA_ROWS` adds 18501.
- Centre County is split by an approximation (`CENTRE_WESTERN_ZIPS`).
- The brochure base URL is hard-coded to the 2026 plan year.
- `--upload` overwrites `regions.json`; DA edits are lost.
- To switch to enGen's list, replace the input and keep the columns.

Migration tooling

Tool	Purpose	Keep after migration?
`importer/`	Import scripts per page template (`import-<template>.js`, about 40), `parsers/` (20 block parsers), `transformers/` (21), `urls-<template>.txt`, esbuild bundles (`*.bundle.js`), `upload-to-da.mjs`	Until the remaining highmark.com pages are migrated
`provider-import/`	Custom converter for providers.highmark.com: `extract.mjs` (Playwright), `build.mjs`, `upload.mjs`, `admin.mjs` (preview, live, index, status), `mark.mjs` (checklist). See its README for a batch end to end.	Until the remaining provider pages are done; `admin.mjs` stays useful
`sidenav/sync-sidenav.mjs`	Rebuilds left-nav blocks from highmark.com's menu (`--apply`)	**No:** it overwrites hand edits; retire at cut-over

Tool	Purpose	Keep after migration?
`breadcrumbs/add-breadcrumbs.mjs`	Adds `Breadcrumbs` and `Breadcrumb Title` metadata from the source's trail	No
`demo/`, `demo-app*`	DA App SDK samples from the boilerplate	No (delete)
`sidekick/config.json`	Legacy Sidekick config (project name still "DA Block Collection")	Fix the name

Importer notes:

- The importer's runner (Adobe's import tooling) isn't in the repo. Edit `import-<template>.js`, then rebuild its bundle with `npx @adobe/aem-import-helper bundle --importjs tools/importer/import-<x>.js`; the runner executes the bundle.
- The runner writes `content/<path>.plain.html` (gitignored). `upload-to-da.mjs` then:
  - makes highmark.com links to live pages relative;
  - restores non-breaking spaces;
  - uploads images into DA dot-folders;
  - uploads and previews the document.
- Re-imports overwrite DA edits. After a batch:
  1. run `build-redirects.mjs --upload`;
  2. run `sync-sidenav.mjs`;
  3. run `add-breadcrumbs.mjs` where needed.

Testing and CI

CI

`.github/workflows/main.yaml` runs on every push: Ubuntu, Node 20, `npm ci`, `npm run lint`. **No tests run in CI** and there is no deploy step; code reaches preview and live through the AEM Code Sync app as soon as a branch is pushed (`<branch>--highmark--adobedrago.aem.page`).

Test suites (`tools/tests/`)

Playwright scripts run with `node tools/tests/<suite>.mjs`. They print PASS / FAIL lines and a summary; screenshots go to the gitignored `tools/tests/shots/`.

`HOST` selects the target:

- the default `http://localhost:3100` is `aem up --port 3100 --no-open`, i.e. your working tree with preview content;
- `HOST=https://main--highmark--adobedrago.aem.page` tests preview;
- `HOST=https://main--highmark--adobedrago.aem.live` tests live (the default of `regression`, `provider-links` and `list-links`).

Suite	Covers
`regression.mjs`	18 key pages at 1280 and 390: console errors, failed requests, broken images, unloaded blocks, horizontal scroll, H1
`search.mjs`, `suggestions.mjs`, `phone-search.mjs`	Header type-ahead, `/search` ranking and paging, suggestions sheet, phone search
`region.mjs`, `shop-chrome.mjs`	ZIP modal, stored selection, location line, region links; shop header and footer per region
`provider-header.mjs`, `provider-pages.mjs`, `provider-content.mjs`, `provider-lists.mjs`	Provider header, landing pages, content pages and articles, list pages
`provider-links.mjs`, `list-links.mjs`	Every link on provider pages and fragments; every list-sheet link through `/redirects`
`a11y.mjs`	axe-core 4.10.2, WCAG 2.1 A/AA plus best practice, at 1280 and 390

Some suites hard-code expected data (`suggestions.mjs`'s `EXPECTED`, `provider-lists.mjs`'s `TOTALS`, a first search result in `search.mjs`) and fail after legitimate content changes; update the expectations with the content.

Environments and Git Workflow

Environment	URL	Purpose
Local	`http://localhost:3000`	Development
Branch	`https://<branch>--highmark--adobedrago.aem.page`	PR testing (code from the branch, preview content)
Preview	`https://main--highmark--adobedrago.aem.page`	Staging: `main` code, previewed content
Live	`https://main--highmark--adobedrago.aem.live`	Production (no custom domain yet)

Workflow:

1. Branch off `main`. Keep branch names short and lowercase: they become part of the preview hostname, which is limited to 63 characters.
2. Run `npm run lint`, then the relevant test suites against `aem up --port 3100`.
3. Open a PR with the branch preview URLs of the pages you changed. The PR template still has the boilerplate's URLs; replace them.
4. Squash-merge and delete the branch. `main` is live as soon as it's merged.

Code changes go live on merge, but **content** needs its own preview and publish. If a change needs new content (a sheet, a fragment, a library doc), publish it with the merge.

Common Development Tasks

Add a block

1. Create `blocks/<name>/<name>.js` exporting `default function decorate(block)`, and `blocks/<name>/<name>.css` scoped to `.<name>`.
2. Build a test page in DA (or under `content/` with `aem up --html-folder content`) using the real content model.
3. Check it at 390, 768, 992 and 1280 against the source design, and run `regression.mjs` on the page.
4. Add a Library entry: a doc under `/docs/library/blocks/<name>` (one section per variant: a `library-metadata` block, then the block) and a row in `/docs/library/blocks.json`; preview both.
5. Lint, PR, merge.

Add a template

1. Create `styles/templates/<name>.css`, with every rule scoped to `body.<name>`.
2. Add `<name>` to `TEMPLATE_STYLES` in `scripts/scripts.js`.
3. If the template needs JS, add a body-class-gated decorator in `loadEager`, like `decorateFaqLinks`.
4. If the left nav is used, add the template to the `body:not(...)` list in `cards-minimal-light-sidenav.css`.

Add a section style

Add `.section.<style>` rules to `styles.css` (global) or the template file (scoped), then document the value for authors.

Add a header or footer fragment

Create the fragment in DA with the section shape its layout expects. If it is a new `nav` path, add its height reservation to `styles.css`. Add it to `FRAGMENTS` in `build-redirects.mjs` if it holds links.

Add analytics or a marketing tool

Load it from `scripts/delayed.js`, never from `scripts.js` or `head.html`. Check it under the CSP and in the Network tab, and confirm Lighthouse isn't affected.

Refresh generated content

Content	Steps
Provider news lists	`build-provider-lists.mjs --upload` → check preview → publish the 5 sheets → `build-redirects.mjs --upload` → compare DA with live → publish `/redirects.json`

Content	Steps
Press releases, events	`build-press-releases-data.mjs --upload` (or events) → check preview → publish
ZIP data	`build-zip-data.mjs --upload` → check the shop on preview (`region.mjs`, `shop-chrome.mjs`) → publish both sheets
After publishing new pages	`build-redirects.mjs --upload` → publish the sheet with the pages → re-index pages published while redirected

Debug a block

1. Check the console and the Network tab for 404s on `/blocks/<name>/<name>.{js,css}`; the folder name must match the block class.
2. Compare the block's `.plain.html` (`<page>.plain.html`) with what the JS expects. Authors' tables arrive as nested `div`s.
3. Check which template stylesheets also style the block.
4. `data-block-status` on the block shows whether it loaded.

Known Issues

Found during the handover analysis on 2026-10-21. Content issues (broken links, missing titles) are tracked in `BACKLOG.md`.

Functional

1. **Forms can't submit.** `form-minimal-light` on `/contact`, `/fraud/fraud-form` and `/newsroom/news-alert` POSTs JSON to `https://www.highmark.com/bin/hmk/genericmailer`. Its CORS preflight returns no `Access-Control-Allow-Origin` header, so browsers block the request, and the endpoint expects a reCAPTCHA token. The base `form.js` reports failures only to the console, so visitors get no message.
2. **The Universal Editor branch breaks page load.** `scripts.js` top-level-awaits `/ue/scripts/ue.js` on `*.ue.da.live` hosts, and `ue/` was deleted from the repo. There, `scripts.js` fails before `loadPage()` runs. Remove the branch, or restore the file if Universal Editor is wanted.
3. **Product shop headers shift the page.** `/shop/fragments/shopx-header-dental` and `-blue-edge-balance` don't match the `/shopx-header` height rule, so they get the corporate reservation (84px / 196.5px instead of 67px / 208px).
4. **Hard-coded sheet limits truncate silently:** ZIP sheets 10,000 rows (3,810 today; enGen's list may be larger), search index 5,000, query index 5,000, provider lists 5,000, press releases 1,000.
5. **`cards-minimal-dark-list` adds one `hr` per cell** under each H3: the H3 loop sits inside the per-cell loop.
6. **The DA Library's Templates entry points at a missing sheet** (`/docs/library/templates.json`).

Fragile coupling

7. **CSS keyed on heading ids.** Rewording these headings silently changes the layout:
   - `#types-of-eating-disorders-and-treatment` (`cards-minimal-dark-withimg.css`);
   - `#employee-health-care-management`, `#highmark-client-resources-for-employers`, `#education-for-employers` (`employer-subpage.css`);
   - `#app-developers-api-support` (`legal.css`);
   - `#highmark-ventures-is-transforming-healthcare`, `#medicare-d-snp-medicaid-and-chip` (`campaign-landing.css`);
   - two in `bright-blue-futures.css`;
   - `#mental-health-resources`, `#eating-disorders` (`hero-minimal-dark-withimg.css`);
   - `#elizabeths-story` (`columns-minimal-dark-overlay.css`);
   - `#dynamic-table` (`table-minimal-dark-compare.css`).
8. **Classes chosen by matching authored text:** `LEADERSHIP_BIO_CLASSES`, `LEGAL_LINK_CLASSES`, the LinkedIn link, and the region-code regex in `decorateProviderContent`, which duplicates the regions sheet.
9. **Positional fragment parsing.** Header and footer sections are read by index; `inheritMenu` takes the other nav's third section.
10. **Content paths hard-coded in code:** the nav paths in `styles.css` and `header.css`, the footer path prefixes, `/modals/zip-county`, the `/shop/*.json` defaults, `/docs/library/icons/filter.svg` and `/search-suggestions.json`.

11. **External hosts in code:** `https://www.highmark.com/search-results.html` (search fallback), `www.highmark.com` treated as same-site in the footer, a `highmark.com` hostname regex for new-tab logic, `fonts.gstatic.com` (Material Icons) and the `da.live` modules.

Performance

12. **Render-blocking font CSS:** the Adobe Fonts stylesheet loads in `head.html`, and template CSS is awaited before first paint.
13. **Breadcrumbs make N+1 requests:** the full query index, then each ancestor's whole page HTML to read one meta tag.
14. **The shop header and footer fetch their fragment twice** (a raw fetch, then `loadFragment`).
15. **The search index is downloaded whole** (about 280 KB compressed today); it won't scale to the full site.

Accessibility

16. Footer accordion headings are click-only `<p>` elements, with no button or `aria-expanded`.
17. Header suggestion options contain links, which fails axe `nested-interactive`.
18. `#alert-banner-title` is a fixed id, so two banners on a page would collide.

Maintenance

19. **Dead code:**
    - `scripts/delayed.js` and `styles/lazy-styles.css` are empty;
    - `scripts/context.js` and `styles/context.css` are unused;
    - `plugins/experimentation/` is never loaded, so `Experiment` metadata does nothing;
    - the `build:json` scripts and husky hook target the deleted `ue/models`;
    - `fonts/roboto-{regular,medium,bold}.woff2` and the `roboto-condensed` face are unused;
    - `zip-county-form` loads `form.css`, which has no effect;
    - `hero.js` is empty;
    - the `press-releases` template value does nothing.
20. **Duplicated code:**
    - an `element()` helper in 8 files;
    - heading-level and icon helpers;
    - "off-site or PDF opens a new tab" in 4 places;
    - three filter-drawer implementations;
    - `cards-minimal-dark-withimg-2` and `-icons` identical to `cards.js`;
    - `table-minimal-dark-compare` nearly identical to `table`.
21. **Boilerplate leftovers:**
    - `package.json` name, repository and homepage (`@adobe/aem-block-collection`);
    - the PR template's URLs;
    - `tools/sidekick/config.json`'s project name;
    - stale block READMEs;
    - `blocks/*/metadata.json` (importer metadata, publicly served).
22. **Tooling gaps:**

    - `tools/importer/upload-to-da.mjs` sends no `Authorization` header, so it gets 401 from a local clone; give it the other tools' `readToken()`;
    - no Node version pin (add `.nvmrc` with `20` and `engines`);
    - CI runs lint only; consider adding `regression.mjs` against the branch preview.
23. **Library docs log two 404s on preview:** they request a `library-metadata` block that doesn't exist. Preview only.

Troubleshooting

Issue	Check
Block not loading	Folder name matches the block class? JS exports `default function decorate`? 404 for its JS or CSS in Network?
Styles not applying	Is a template stylesheet overriding the block? Is the template in `TEMPLATE_STYLES`? Specificity?
Page waits blank, then appears	Template CSS failed to load: `body.appear` waits for it
Content not updating	Was the document previewed (not just saved in DA)? Hard refresh; the CDN can take a minute or two
A published page redirects to highmark.com	A `/redirects` row covers its path; regenerate and publish the sheet
Page missing from search or breadcrumbs	Not indexed: it was published while a redirect covered it. Re-index it.
Header jumps on load	The `nav` path has no height reservation in `styles.css`
Header or footer layout broken	Fragment sections reordered, or a section count changed (the layouts are positional)
Region links missing on a shop page	No stored ZIP yet, or the `regions` column is empty for that region; check `shop-zip-county` in localStorage
`about:error` images or a broken share image	The `<img src>` or `Image` metadata isn't reachable when previewing
Tool gets 401	DA token missing or expired: `npx github:adobe-rnd/da-auth-helper token`
Lint fails on Windows	CRLF line endings: set VS Code `files.eol` to `\n` and re-save

Resources and Support

Documentation

Resource	URL
AEM Edge Delivery Services documentation	`https://www.aem.live/docs/`
Developer tutorial	`https://www.aem.live/developer/tutorial`
Block Collection	`https://www.aem.live/developer/block-collection`
Keeping it 100 (loading phases)	`https://www.aem.live/developer/keeping-it-100`
Development good practices	`https://www.aem.live/docs/dev-collab-and-good-practices`
Admin API	`https://www.aem.live/docs/admin.html`
Document Authoring	`https://docs.da.live/`
Repo notes	`AGENTS.md` (working notes), `BACKLOG.md` (open issues), `MIGRATION-CHECKLIST.md` (page status), `tools/provider-import/README.md`, `tools/tests/README.md`

Support contacts

Need	Contact
Site configuration and access	[Site administrator: name, email]
Code and tooling questions	[Development lead: name, email]
Adobe support	[Adobe account contact]